# The mean value property

# Where we stand

Last lecture: $\Omega$ a finite union of grid squares, $g$ linear on each boundary edge of the coarsest grid, $U_n$ the finite element solution on the grid of size $\varepsilon_n = 2^{-n}\varepsilon_0$. We proved:

- $U_n \to u$ and $\nabla U_n \to \nabla u$ in $L^2(\Omega)$, and $a(u, \eta) = 0$ for every smooth $\eta$ vanishing on $\partial\Omega$;

We aim to show today that $u$ is smooth and $\Delta u = 0$ pointwise.

# The mean value property

## Theorem (mean value property)

Let $h$ be $C^2$ and harmonic on $\Omega$, and let the closed disk $|z - z_0| \le r$ lie in $\Omega$. Then $h(z_0) = \dfrac{1}{2\pi} \displaystyle\int_0^{2\pi} h(z_0 + re^{i\theta})\, d\theta$.

**Proof.** Let $M(t)$ be the average of $h$ over the circle $C_t$ of radius $t \le r$ about $z_0$, and $D_t$ the disk inside. Differentiate under the integral. By the chain rule, $\partial_t \big[h(z_0 + te^{i\theta})\big] = \nabla h \cdot e^{i\theta}$, and $n = e^{i\theta}$ is the outward normal to $C_t$, where $ds = t\, d\theta$:

$$M'(t) \;=\; \frac{1}{2\pi} \int_0^{2\pi} \nabla h \cdot n \, d\theta \;=\; \frac{1}{2\pi t} \oint_{C_t} \nabla h \cdot n \, ds \;=\; \frac{1}{2\pi t} \iint_{D_t} \Delta h \, dA \;=\; 0.$$

So $M$ is constant, and $M(t) \to h(z_0)$ as $t \to 0$ by continuity. $\square$

# The strong maximum principle

## Theorem (strong maximum principle)

Let $\Omega$ be open and **connected**, and let $h$ be $C^2$ and harmonic on $\Omega$. If $h(z_0) = \sup_\Omega h$ at some point $z_0 \in \Omega$, then $h$ is constant on $\Omega$.

**Proof.** Put $M := h(z_0)$ and $S := \{z \in \Omega : h(z) = M\}$. Then $S$ is not empty, and it is closed in $\Omega$ because $h$ is continuous. We show that $S$ is open; since $\Omega$ is connected, it follows that $S = \Omega$.

Let $z_1 \in S$ be any point of $S$, and let the closed disk of radius $r$ about $z_1$ lie in $\Omega$. For each $t \leq r$, the average of $h$ over the circle $C_t$ about $z_1$ is $h(z_1) = M$, while $h \leq M$ on $C_t$. If $h < M$ at one point of $C_t$ then, by continuity, $h < M$ on an arc, and the average would be $< M$. So $h = M$ on every $C_t$: the disk lies in $S$. $\square$

# The maximum principle, and uniqueness

## Corollary (maximum principle)

Let $\Omega$ be bounded and open, and $h \in C^2(\Omega) \cap C(\overline{\Omega})$ harmonic. Then $\max_{\overline{\Omega}} h = \max_{\partial\Omega} h$ and $\min_{\overline{\Omega}} h = \min_{\partial\Omega} h$.

**Proof.** $h$ attains its maximum $M$ over the compact set $\overline{\Omega}$ at some $z_0$. If $z_0 \in \Omega$, apply the theorem on the connected component $\Omega'$ of $z_0$: $h \equiv M$ on $\Omega'$, hence on $\partial\Omega'$ by continuity, and $\partial\Omega'$ is a nonempty part of $\partial\Omega$. For the minimum, take $-h$. $\square$

## Corollary (stability)

If $h_1, h_2 \in C^2(\Omega) \cap C(\overline{\Omega})$ are harmonic, then $\max_{\overline{\Omega}} |h_1 - h_2| = \max_{\partial\Omega} |h_1 - h_2|$. In particular, if $h_1 = h_2$ on $\partial\Omega$ then $h_1 = h_2$ on $\Omega$.

Apply the maximum principle to $h_1 - h_2$.

# Averaging over disks

Fix a smooth $\chi \geq 0$ on the plane, **radial**, zero outside the unit disk, with $\iint \chi = 1$. Put $\chi_\delta(z) := \delta^{-2}\chi(z/\delta)$: zero outside the disk of radius $\delta$, integral 1. Let $\Omega_\delta$ be the set of $z \in \Omega$ whose closed disk of radius $\delta$ lies in $\Omega$. For $v$ on $\Omega$ and $z \in \Omega_\delta$, the convolution $(v * \chi_\delta)(z) := \iint v(\zeta)\,\chi_\delta(z-\zeta)\,dA(\zeta)$ is a weighted average of $v$ over that disk.

## Lemma (averaging)

Let $v$ be continuous on $\Omega$ and have the **mean value property**: $v(z)$ is the average of $v$ over every circle about $z$ whose closed disk lies in $\Omega$. Then $v * \chi_\delta = v$ on $\Omega_\delta$.

**Proof.** Write $\chi_\delta(\zeta) = k(|\zeta|)$, so that $\int_0^\delta k(t)\,2\pi t\,dt = 1$. In polar coordinates about $z$, circle by circle,

$$(v * \chi_\delta)(z) = \int_0^\delta k(t)\Big(\int_0^{2\pi} v(z + te^{i\theta})\,d\theta\Big)\,t\,dt = v(z)\int_0^\delta k(t)\,2\pi t\,dt = v(z). \;\square$$

# Weyl's lemma

## Weyl's lemma

Let $u \in L^2(\Omega)$ satisfy $\iint_\Omega u\,\Delta\eta\,dA = 0$ for every **test function** $\eta$: a smooth function on $\Omega$ that vanishes outside a compact subset of $\Omega$, hence near $\partial\Omega$. Then $u$ agrees almost everywhere with a smooth harmonic function.

**What the hypothesis means.** Suppose for a moment that $u$ is $C^2$. Integrate by parts twice; there are no boundary terms, because $\eta$ vanishes near $\partial\Omega$:

$$\iint_\Omega u\,\Delta\eta\,dA \;=\; \iint_\Omega (\Delta u)\,\eta\,dA.$$

This vanishes for every test function $\eta$ exactly when $\Delta u = 0$.

For $u$ only in $L^2$, only the left side makes sense. The hypothesis is "$\Delta u = 0$" **with both derivatives moved onto** $\eta$, and the lemma says that even this weakest form of the equation forces $u$ to be smooth.

# Weyl's lemma applies to our $u$

**What Lecture 3 proved.** $U_n$ has a limit $u$ in $L^2$, and $\nabla U_n$ has a limit $(p, q)$ in $L^2$. This limiting gradient satisfies $\iint (p\,\partial_x\eta + q\,\partial_y\eta)\,dA = 0$ for every test function $\eta$.

**Integrate by parts, for each $n$.** $U_n$ is continuous, and smooth on each grid square. Apply the divergence theorem to $U_n\nabla\eta$ on each square and add up. On the grid lines the contributions of neighbouring squares cancel, because $U_n$ is continuous. Near $\partial\Omega$ there are none, because $\eta$ vanishes there. So

$$\iint \nabla U_n \cdot \nabla\eta \; dA \;=\; -\iint U_n\,\Delta\eta \; dA.$$

**Let $n \to \infty$.** The left side tends to $\iint (p\,\partial_x\eta + q\,\partial_y\eta) = 0$, and the right side to $-\iint u\,\Delta\eta$. So $\iint u\,\Delta\eta\,dA = 0$ for every test function $\eta$: Weyl's lemma applies to $u$.

# Weyl's lemma: smoothing

Put $u_\delta := u * \chi_\delta$, defined on $\Omega_\delta$, the set of $z$ whose closed disk of radius $\delta$ lies in $\Omega$.

- $u_\delta$ **is smooth.** Differentiate under the integral sign: every derivative falls on $\chi_\delta$.
- $u_\delta$ **is harmonic.** Fix $z \in \Omega_\delta$. The function $\eta(\zeta) := \chi_\delta(z - \zeta)$ is a test function: it is smooth, and it vanishes outside the closed disk of radius $\delta$ about $z$, which lies in $\Omega$. Also $\Delta_z$ and $\Delta_\zeta$ agree on a function of $z - \zeta$. So

$$\Delta u_\delta(z) \;=\; \iint u(\zeta)\, \Delta_z\big[\chi_\delta(z-\zeta)\big]\, dA(\zeta) \;=\; \iint u\, \Delta\eta \; dA \;=\; 0.$$

## Standard fact (smoothing)

If $u \in L^2(\Omega)$, then $u * \chi_\delta \to u$ in $L^2(K)$ as $\delta \to 0$, for every compact $K \subset \Omega$.

# Weyl's lemma: smoothing twice

**Two facts.** (1) $u_\delta = u * \chi_\delta$ is harmonic (previous slide). (2) Smoothing a harmonic function does nothing: $h * \chi_{\delta'} = h$ (the averaging lemma, slide 6).

**Smooth $u$ twice.** By (1) and (2), smoothing $u_\delta$ again does nothing. But the order of two smoothings does not matter (Fubini). So, wherever both sides are defined,

$$u_\delta \;=\; (u * \chi_\delta) * \chi_{\delta'} \;=\; (u * \chi_{\delta'}) * \chi_\delta \;=\; u_{\delta'} * \chi_\delta \;=\; u_{\delta'}.$$

**All the $u_\delta$ are one and the same function:** there is a single smooth harmonic $v$ on $\Omega$ with $u_\delta = v$ at all points farther than $2\delta$ from $\partial\Omega$.

**And $v = u$.** On a compact $K \subset \Omega$, the standard fact says $u_\delta \to u$ in $L^2(K)$. But $u_\delta = v$ on $K$ once $\delta$ is small. So $u = v$ almost everywhere. $\square$ From now on $u$ denotes $v$: we changed $u$ only on a set of area zero.

# The converse of the mean value property

## Theorem (converse of the mean value property)

A continuous function $v$ on $\Omega$ with the mean value property is smooth and harmonic.

**Proof: $v$ is smooth.** By the averaging lemma (slide 6), $v = v * \chi_\delta$ on $\Omega_\delta$. In $(v * \chi_\delta)(z) = \iint v(\zeta)\, \chi_\delta(z - \zeta)\, dA(\zeta)$ the variable $z$ appears only inside $\chi_\delta$. Differentiate under the integral sign: every derivative falls on $\chi_\delta$, which is smooth, and $v$ only needs to be continuous. So $v$ is smooth on $\Omega_\delta$ for every $\delta$, that is, on $\Omega$.

**Proof: $v$ is harmonic.** Let $M(t)$ be the average of $v$ over the circle of radius $t$ about $z_0$. For smooth $v$ the proof of the mean value property gave $M'(t) = \frac{1}{2\pi t} \iint_{D_t} \Delta v\, dA$. Here $M(t) = v(z_0)$ is constant, so $\iint_{D_t} \Delta v\, dA = 0$ for every small $t$. Divide by the area $\pi t^2$ and let $t \to 0$: since $\Delta v$ is continuous, $\Delta v(z_0) = 0$. $\square$

# Uniform limits of harmonic functions

## Corollary (uniform limits)

Let $h_n$ be harmonic on $\Omega$ and let $h_n \to v$ uniformly on every compact subset of $\Omega$. Then $v$ is smooth and harmonic.

**Proof.** First, $v$ is continuous: near each point it is a uniform limit of continuous functions.

Next, $v$ has the mean value property. Let the closed disk $D$ of radius $r$ about $z_0$ lie in $\Omega$. Each $h_n$ is harmonic, so $h_n(z_0) = \frac{1}{2\pi}\int_0^{2\pi} h_n(z_0 + re^{i\theta})\,d\theta$. Let $n \to \infty$. The left side tends to $v(z_0)$. On the right, $D$ is compact, so the convergence is uniform on the circle:

$$\left|\frac{1}{2\pi}\int_0^{2\pi} h_n(z_0 + re^{i\theta})\,d\theta \;-\; \frac{1}{2\pi}\int_0^{2\pi} v(z_0 + re^{i\theta})\,d\theta\right| \;\le\; \max_D |h_n - v| \;\longrightarrow\; 0.$$

Hence $v(z_0)$ is the average of $v$ over the circle. By the converse of the mean value property (slide 11), $v$ is smooth and harmonic. □

# Uniform limits: what just happened

A uniform limit of smooth functions is usually **not** smooth: polynomials converge uniformly on $[-1, 1]$ to $|x|$.

Harmonic functions escape this for two reasons, and it is worth keeping them apart.

1. **Harmonic = continuous + mean value property**, and the right side has no derivative in it. A statement about values and integrals survives uniform convergence; a statement about derivatives would not.
2. The smoothness of the limit is **not inherited** from the $h_n$. It is produced afresh, by the identity $v = v * \chi_\delta$: an average of a continuous function against a smooth weight is smooth.

Holomorphic functions will behave in the same way, for the same reason: Cauchy's formula is a statement about values and integrals.